Brief Overview of Logistic Regression

- **Optimization in Logistic Regression**

Learning Objectives

- Explain two different optimization algorithms: Gradient Descent and Newton-Raphson

- Apply the Newton-Raphson algorithm to estimate parameters in logistic regression

Logistic Regression

- **Data**: $(Y_i, x_{i1}, \cdots, x_{i,p-1})$ for $i = 1, \cdots, n$, where $Y_i \in \{0, 1\}$ is the class label.

- **Logistic Regression Model** is defined as two components:
  - $P(Y_i = 1) = \pi_i$ and $P(Y_i = 0) = 1 - \pi_i$
  - $\log \frac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_{p-1} x_{i,p-1}$

- **Question:** How to estimate the β_i's in the model?

MLE in Logistic Regression

- **Maximum Likelihood Estimator** (MLE) $\hat{\boldsymbol{\beta}}$ of the $\boldsymbol{\beta}_i$'s can be found by maximizing the likelihood function

$$\boldsymbol{L}(\boldsymbol{\beta}) = \prod_{i=1}^{n} \frac{e^{y_i(\beta_0+\beta_1 x_{i1}+\beta_2 x_{i2}+\cdots+\beta_{p-1} x_{i,p-1})}}{1 + e^{\beta_0+\beta_1 x_{i1}+\beta_2 x_{i2}+\cdots+\beta_{p-1} x_{i,p-1}}}$$

- **Challenges:**
    - ➢ In general there are no explicit solutions to the MLE $\hat{\boldsymbol{\beta}}$.
    - ➢ We need to rely the computers to apply an efficient optimization algorithm to numerically find the MLE $\hat{\boldsymbol{\beta}}$

Optimization Problem

In statistics and machine learning, we often face the optimization problem

$$\hat{\theta} = \underset{\theta \in \Theta \subset R^p}{\operatorname{argmin}}\, h(\theta)$$

- Often $h(\theta)$ is smooth, and we want to solve $h'(\theta) = 0$.
- **Iterative Method:** one widely used optimization algorithm:
  - find a sequence of $\theta^{(i)}$ values until convergence to $\hat{\theta}$
  - When convergence, we have $h'\left(\hat{\theta}\right) = 0$

Optimization Algorithms

Two basic iterative algorithms to solve $\hat{\theta} = \underset{\theta}{\operatorname{argmin}}\, h(\theta)$:

1. **Gradient Descent Algorithm: widely used in Machine Learning:**

$$\boldsymbol{\theta_{new} = \theta_{old} - \lambda\, h'(\theta_{old})}$$

   where λ is the so-called learning rate.

2. **Newton-Raphson Method: very popular in statistics:**

$$\boldsymbol{\theta_{new} = \theta_{old} - [h''(\theta_{old})]^{-1} h'(\theta_{old})}$$

Newton-Raphson Method

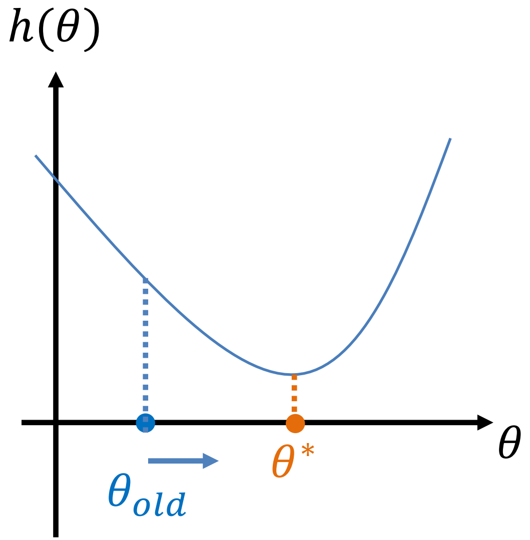
<u>**Mathematical idea**</u> behind Newton-Raphson method for the optimization problem

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}}\, h(\theta)$$

- If we start at θ_{old} and want to update to

$$\theta_{new} = \theta_{old} + \epsilon,$$

how to choose $\boldsymbol{\epsilon}$ so that $\boldsymbol{h'(\theta_{new}) = 0}$?



- Taylor series expansion of $h(\theta)$ on θ_{old}!

Taylor Expansion Consideration

- When θ is one-dim, by **Taylor series expansion,**

$$h'(\theta_{old} + \epsilon) \approx h'(\theta_{old}) + \epsilon h''(\theta_{old}).$$

- Setting this to 0 yields that

$$\boldsymbol{\epsilon \approx -\frac{h'(\theta_{old})}{h''(\theta_{old})}}$$

- The idea is identical for high-dim θ. This leads to Newton-Raphson Method:

$$\boldsymbol{\theta_{new} = \theta_{old} - [h''(\theta_{old})]^{-1} h'(\theta_{old})}$$

(update until θ converges)

Newton-Raphson Method in Statistics

Why is Newton-Raphson Method very popular in Statistics?

- When convergence, Newton-Raphson Method provides two values:

$$\hat{\boldsymbol{\theta}} \quad \text{and} \quad \hat{\boldsymbol{V}} = -[\boldsymbol{h}''(\hat{\boldsymbol{\theta}})]^{-1}$$

- When $\boldsymbol{h}(\boldsymbol{\theta})$ is the log-likelihood function in statistics, the maximum likelihood estimator (MLE) is $\hat{\boldsymbol{\theta}}$.

  - By the asymptotic theory of MLE, $\dfrac{\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}}{\sqrt{\hat{\boldsymbol{V}}}} \sim N(0,1)$

    and thus 95% confidence interval on $\boldsymbol{\theta}$ is $\hat{\boldsymbol{\theta}} \pm \mathbf{1.96}\sqrt{\hat{\boldsymbol{V}}}$

MLE of Logistic Regression

- In logistic regression, the log-likelihood function (in vector notation)

$$\log L(\beta) = \sum_{i=1}^{n} [y_i \beta^\top x_i - \log(1 + e^{\beta^\top x_i})]$$

- **First-order derivatives**

$$\frac{\partial \log L(\beta)}{\partial \beta} = \sum_{i=1}^{n} \left[y_i x_i - \frac{\boldsymbol{e^{\beta^\top x_i}}}{\mathbf{1} + \boldsymbol{e^{\beta^\top x_i}}} x_i \right] = \sum_{i=1}^{n} (y_i - \boldsymbol{\pi_i}) x_i$$

- **Second-order derivatives**

$$\frac{\partial^2 \log L(\beta)}{\partial \beta \partial \beta^\top} = -\sum_{i=1}^{n} x_i \frac{e^{\beta^\top x_i}\left(1 + e^{\beta^\top x_i}\right) - e^{\beta^\top x_i} \cdot e^{\beta^\top x_i}}{\left(1 + e^{\beta^\top x_i}\right)^2} x_i^\top = -\sum_{i=1}^{n} x_i \pi_i (1 - \pi_i) x_i^\top$$

Newton-Raphson Method

- Let $\pi = \begin{pmatrix} \pi_1 \\ \cdots \\ \pi_n \end{pmatrix}$; $W = \begin{pmatrix} \pi_1(1-\pi_1) & & \\ & \ddots & \\ & & \pi_n(1-\pi_n) \end{pmatrix}$ --- both depend on β_{old}
- We have $\boldsymbol{h'(\beta) = X^T(Y - \pi)}$ and $\boldsymbol{h''(\beta) = -X^T W X}$.
- Applying Newton-Raphson Method to the MLE of logistics regression,

$$\begin{aligned} \boldsymbol{\beta_{new}} &= \boldsymbol{\beta_{old} - [h''(\beta_{old})]^{-1} h'(\beta_{old})} \\ &= \boldsymbol{\beta_{old}} + \left(\boldsymbol{X}^\top \boldsymbol{W} \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top (\boldsymbol{Y} - \boldsymbol{\pi}) \\ &= \left(\boldsymbol{X}^\top \boldsymbol{W} \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{W} \boxed{\left[\boldsymbol{X}\boldsymbol{\beta_{old}} + \boldsymbol{W}^{-1}(\boldsymbol{Y} - \boldsymbol{\pi})\right]} \\ &= \left(\boldsymbol{X}^\top \boldsymbol{W} \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{W} \boldsymbol{Z}, \end{aligned}$$

where Z is adjusted response. This is weighted least squares.

Algorithm for MLE in logistics

Newton-Raphson algorithm to compute the MLE in logistic regression model:

- Initialize $\beta_{init} = 0$
- Given β_{old},
  - **compute three new variables**:

$$\widehat{\boldsymbol{\pi}}_{\boldsymbol{i}} = \frac{e^{\beta_{old}^{\top}\boldsymbol{x}_{\boldsymbol{i}}}}{1 + e^{\beta_{old}^{\top}\boldsymbol{x}_{\boldsymbol{i}}}}; \quad \boldsymbol{w}_{\boldsymbol{i}} = \hat{\pi}_i(1 - \hat{\pi}_i); \quad \boldsymbol{Z}_{\boldsymbol{i}} = \beta_{old}^{\top}\boldsymbol{x}_{\boldsymbol{i}} + \frac{Y_i - \hat{\pi}_i}{\hat{\pi}_i(1 - \hat{\pi}_i)}$$

  - **Conduct weighted least squares**:

$$\beta_{new} \leftarrow \underset{\beta}{\operatorname{argmin}} \left[(Z - X\beta)^{\top} W (Z - X\beta)\right]$$

- Repeat the second step until converge.

[Remarks: the algorithm fails only when $\hat{\pi}_i = 0$ or 1, i.e., with perfect fits.]

Summary

- Two optimization algorithms:
  - Gradient Descent
  - Newton-Raphson

- Application of Newton-Raphson Method to find the MLE of the logistic regression model